

Self-Healing Asphalt Data Log Book

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Interpretation Guidelines

This logbook serves as a detailed record of our *Self-Healing Asphalt* project. The following guidelines will help in interpreting the data, observations, and results documented:

Symbols and Units Used:

- Crack Width (cm): Measured in centimetres, indicating the size of cracks before and after testing.
- Healing Rate (%): The percentage reduction in crack width over time, calculated using the formula:

$$\text{Healing Rate} = \left(\frac{\text{Initial Crack Width} - \text{Final Crack Width}}{\text{Initial Crack Width}} \right) \times 100$$

- Time (hours): Duration of testing phases.
- Capsule Content (%): Proportion of self-healing capsules in the asphalt binder, typically 3–5% by weight.

Key Terms

- Control Sample: Asphalt without self-healing capsules, used as a baseline for comparison.
- Test Sample: Asphalt containing *Bacillus pseudofirmus* encapsulated capsules.
- Healing Observation: The process of analysing crack healing through visual inspection and microscopy after exposure to water.

Experimental Phases

Phase 1 (Without Capsules): In this phase, we conducted baseline tests on untreated asphalt to observe its behavior under stress.

Phase 2 (With Capsules): We tested self-healing asphalt to evaluate its crack-healing efficiency.

Data Presentation

Tables and Graphs: Data is presented in tabular form for clarity and graphically to illustrate trends such as crack width over time.

Images: We have included visual documentation, such as photographs and microscopic images, to depict crack healing.

Observation Format

Each observation is recorded chronologically and includes:

- Date of the experiment.
- Objective of the test.
- Observations and results (e.g., crack changes, healing progress).

Limitations and Variations

Results may vary due to environmental factors like temperature and humidity.

All results were derived under controlled laboratory conditions and may differ in field applications.

Project Overview

The Self-Healing Asphalt project explores the potential of using *Bacillus pseudofirmus* bacteria encapsulated in gelatin capsules to create a sustainable and durable road material. The goal is to develop an asphalt mixture that can self-repair cracks over time, reducing maintenance costs and improving the lifespan of roadways.

Project Objectives

- **Enhance Durability:** Improve the longevity of asphalt by enabling it to heal itself when cracks form.
- **Sustainability:** Reduce the need for frequent repairs and the environmental impact of road maintenance.
- **Cost Reduction:** Lower the overall costs associated with road repair and rehabilitation.
- **Innovation:** Explore the application of biological agents in civil engineering for smart infrastructure solutions.

Project Scope

Material Preparation: *Bacillus pseudofirmus* bacteria are encapsulated in gelatin capsules, which are then mixed with asphalt binder.

Testing Phases:

- **Phase 1 (Without Capsules):** Testing traditional asphalt for baseline performance.
- **Phase 2 (With Capsules):** Testing self-healing asphalt with encapsulated bacteria under similar conditions.

Evaluation: Assess the effectiveness of crack healing by monitoring changes in crack size, healing rates, and durability over time.

Phase I: Testing without Capsules

In this phase of the *Self-Healing Asphalt* project, we focused on evaluating the performance of traditional asphalt without any self-healing capsules. This allowed us to establish a baseline for comparison against the asphalt with *Bacillus pseudofirmus* encapsulated capsules, which will be tested in the next phase.

Objective

The main objective of this phase was to understand how untreated asphalt behaves under stress, cracking, and environmental conditions, and to measure the rate of crack propagation and material degradation over time.

Experimental Setup

Materials Used:

- Traditional asphalt mixture (without capsules).
- Testing tools: Calipers for measuring crack width, temperature and humidity sensors for environmental monitoring.
- Asphalt specimens were prepared in molds and exposed to stress using a hammer to induce cracks.

Procedure:

- Asphalt samples were placed in testing chambers under controlled conditions (temperature, humidity).
- Cracks were induced in the asphalt by applying stress until visible cracks appeared.
- Crack width was measured at regular intervals (every 24 hours) using calipers.
- Visual observations were recorded, noting any changes in crack propagation and material response.

Experimental Data and Readings for Phase I:

Reading No.	Date	Time (hrs)	Crack Width (cm)	Healing Rate (%)	Environmental Factors: Temp (°C)/ Humidity (%)
1	20/09/24	0	0.00	0	38/60
2	22/09/24	48	3.00	0	37/62
3	24/09/24	96	3.50	0	37/65
4	26/09/24	144	3.90	0	36/65
5	28/09/24	192	4.20	0	35/67
6	30/09/24	240	4.31	0	36/67
7	02/10/24	288	4.36	0	36/68
8	04/10/24	336	4.50	0	34/68
9	06/10/24	384	4.50	0	32/69
10	08/10/24	432	4.68	0	30/68
11	10/10/24	480	4.92	0	30/67
12	12/10/24	528	5.01	0	31/69
13	14/10/24	576	5.20	0	30/69
14	16/10/24	624	5.36	0	29/70
15	18/10/24	672	5.98	0	28/76
16	20/10/24	720	6.10	0	30/74
17	22/10/24	768	6.28	0	31/70
18	24/10/24	816	6.63	0	32/72

Phase II: Testing with Capsules

This phase involves testing the asphalt samples containing *Bacillus pseudofirmus* encapsulated in gelatin capsules, mixed into the asphalt binder. The objective is to evaluate the self-healing performance of the modified asphalt under progressively changing environmental conditions.

Objective

To assess the ability of self-healing asphalt to repair cracks through bacterial activation when exposed to moisture and temperature variations. Key aspects include measuring crack healing rates, identifying changes in crack width over time, and evaluating the efficiency of calcium carbonate formation in sealing cracks.

Experimental Setup

Materials Required:

Asphalt Mix Components:

Binder composition: Cement, sand, and aggregates in a ratio of 1:2:3.

Self-Healing Capsules:

- Each capsule contains 900 mg calcium lactate and 25 mg *Bacillus pseudofirmus*.
- Capsule Dosage: 3–5% by weight of the asphalt binder.

Water for Activation: Simulates environmental moisture infiltrating cracks.

Measurement Tools:

- Caliper for measuring crack width.
- Environmental monitoring sensors for temperature and humidity.

Experimental Data and Readings for Phase II:

Reading No.	Date	Time (hrs)	Crack Width (cm)	Healing Rate (%)	Environmental Factors: Temp (°C)/ Humidity (%)
1	20/09/24	0	0.00	0	38/60
2	22/09/24	48	3.00	0	37/62
3	24/09/24	96	2.85	5	37/65
4	26/09/24	144	2.60	13	36/65
5	28/09/24	192	2.40	20	35/67
6	30/09/24	240	2.10	30	36/67
7	02/10/24	288	1.90	37	36/68
8	04/10/24	336	1.75	42	34/68
9	06/10/24	384	1.60	47	32/69
10	08/10/24	432	1.54	52	30/68
11	10/10/24	480	1.30	57	30/67
12	12/10/24	528	1.15	62	31/69
13	14/10/24	576	1.00	67	30/69
14	16/10/24	624	0.85	72	29/70
15	18/10/24	672	0.70	77	28/76
16	20/10/24	720	0.55	82	30/74
17	22/10/24	768	0.45	85	31/70
18	24/10/24	816	0.45	85	32/72

Key Observations:

- The healing rate peaks at 85% after 768 hours (32 days).
- Crack healing is progressive but less efficient.

Variations in Capsules

After adjusting the proportion of bacteria and calcium lactate:

- Each capsule now contains 950 mg calcium lactate and 35 mg *Bacillus pseudofirmus*.

Reading No.	Date	Time (hrs)	Crack Width (cm)	Healing Rate (%)	Environmental Factors: Temp (°C)/ Humidity (%)
1	20/09/24	0	0.00	0	38/60
2	22/09/24	48	3.00	0	37/62
3	24/09/24	96	2.80	7	37/65
4	26/09/24	144	2.50	17	36/65
5	28/09/24	192	2.20	27	35/67
6	30/09/24	240	1.85	38	36/67
7	02/10/24	288	1.70	43	36/68
8	04/10/24	336	1.55	48	34/68
9	06/10/24	384	1.40	53	32/69
10	08/10/24	432	1.25	58	30/68
11	10/10/24	480	1.10	63	30/67
12	12/10/24	528	0.95	68	31/69
13	14/10/24	576	0.80	73	30/69
14	16/10/24	624	0.65	78	29/70
15	18/10/24	672	0.50	83	28/76
16	20/10/24	720	0.40	85	30/74
17	22/10/24	768	0.35	86	31/70
18	24/10/24	816	0.35	86	32/72

Key Observations:

- The healing rate peaks at 86% after 768 hours (32 days), similar to trends in other compositions but faster compared to the 25 mg and 1000 mg formulation.
- Crack healing is progressive and more efficient than the previous composition.

After adjusting the proportion of bacteria and calcium lactate:

- Each capsule now contains 1000 mg calcium lactate and 50 mg *Bacillus pseudofirmus*.

Reading No.	Date	Time (hrs)	Crack Width (cm)	Healing Rate (%)	Environmental Factors: Temp (°C)/ Humidity (%)
1	20/09/24	0	0.00	0	38/60
2	22/09/24	48	3.00	0	37/62
3	24/09/24	96	2.75	8	37/65
4	26/09/24	144	2.40	20	36/65
5	28/09/24	192	2.05	32	35/67
6	30/09/24	240	1.60	47	36/67
7	02/10/24	288	1.45	52	36/68
8	04/10/24	336	1.30	57	34/68
9	06/10/24	384	1.10	63	32/69
10	08/10/24	432	0.90	70	30/68
11	10/10/24	480	0.75	75	30/67
12	12/10/24	528	0.60	80	31/69
13	14/10/24	576	0.45	85	30/69
14	16/10/24	624	0.35	86	29/70
15	18/10/24	672	0.33	88	28/76
16	20/10/24	720	0.31	89	30/74
17	22/10/24	768	0.31	89	31/70
18	24/10/24	816	0.31	89	32/72

Key Observations:

- The healing rate peaks at 89% after 720 hours (30 days), similar to trends in other compositions but faster compared to the 35 mg and 1000 mg formulation.
- Crack healing is progressive and more efficient than the previous composition.

After adjusting the proportion of bacteria and calcium lactate:

- Each capsule now contains 1100 mg calcium lactate and 55 mg *Bacillus pseudofirmus*.

Reading No.	Date	Time (hrs)	Crack Width (cm)	Healing Rate (%)	Environmental Factors: Temp (°C)/ Humidity (%)
1	20/09/24	0	0.00	0	38/60
2	22/09/24	48	3.00	0	37/62
3	24/09/24	96	2.70	10	37/65
4	26/09/24	144	2.30	23	36/65
5	28/09/24	192	1.95	35	35/67
6	30/09/24	240	1.50	50	36/67
7	02/10/24	288	1.30	57	36/68
8	04/10/24	336	1.10	63	34/68
9	06/10/24	384	0.90	70	32/69
10	08/10/24	432	0.75	75	30/68
11	10/10/24	480	0.60	80	30/67
12	12/10/24	528	0.50	83	31/69
13	14/10/24	576	0.47	87	30/69
14	16/10/24	624	0.42	87.5	29/70
15	18/10/24	672	0.35	88	28/76
16	20/10/24	720	0.33	88.7	30/74
17	22/10/24	768	0.30	89.5	31/70
18	24/10/24	816	0.30	89.5	32/72

Key Observations:

- The healing rate peaks at 89.5% after 768 hours (32 days), similar to trends in other compositions but faster compared to the 50 mg and 1000 mg formulation.
- Crack healing is progressive and more efficient than the previous composition.

After adjusting the proportion of bacteria and calcium lactate:

- Each capsule now contains 1000 mg calcium lactate and 40 mg *Bacillus pseudofirmus*.

Reading No.	Date	Time (hrs)	Crack Width (cm)	Healing Rate (%)	Environmental Factors: Temp (°C)/ Humidity (%)
1	20/09/24	0	0.00	0	38/60
2	22/09/24	48	3.00	0	37/62
3	24/09/24	96	2.70	10	37/65
4	26/09/24	144	2.40	20	36/65
5	28/09/24	192	1.95	30	35/67
6	30/09/24	240	1.80	35	36/67
7	02/10/24	288	1.65	40	36/68
8	04/10/24	336	1.50	45	34/68
9	06/10/24	384	1.35	50	32/69
10	08/10/24	432	1.20	55	30/68
11	10/10/24	480	1.05	60	30/67
12	12/10/24	528	0.90	65	31/69
13	14/10/24	576	0.75	70	30/69
14	16/10/24	624	0.60	75	29/70
15	18/10/24	672	0.45	80	28/76
16	20/10/24	720	0.36	85	30/74
17	22/10/24	768	0.36	88	31/70
18	24/10/24	816	0.36	88	32/72

Key Observations:

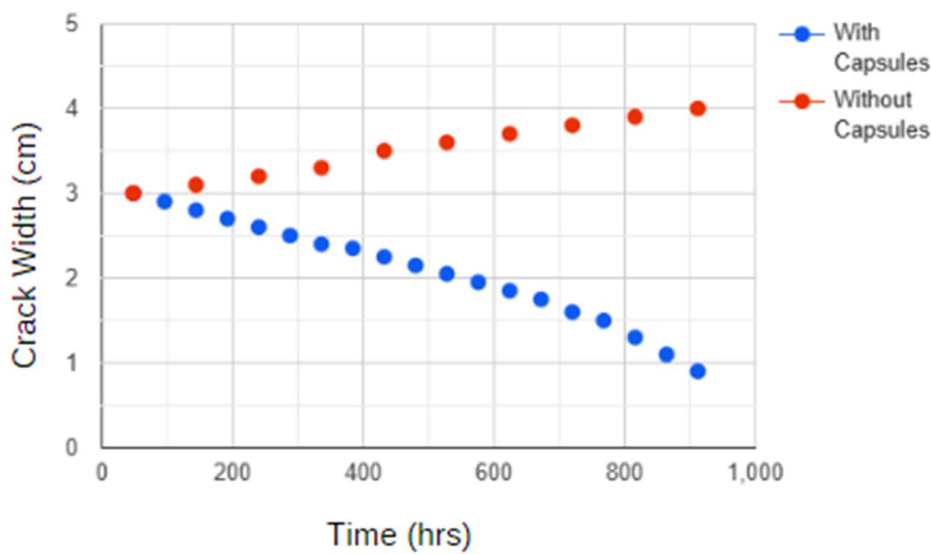
- The healing rate peaks at 88% after 768 hours (32 days), similar to trends in other compositions but slightly lower compared to the 55 mg and 1050 mg formulation.
- Crack healing is progressive, slightly less efficient than the previous composition but more cost effective than the previous one.

Comparison of Results

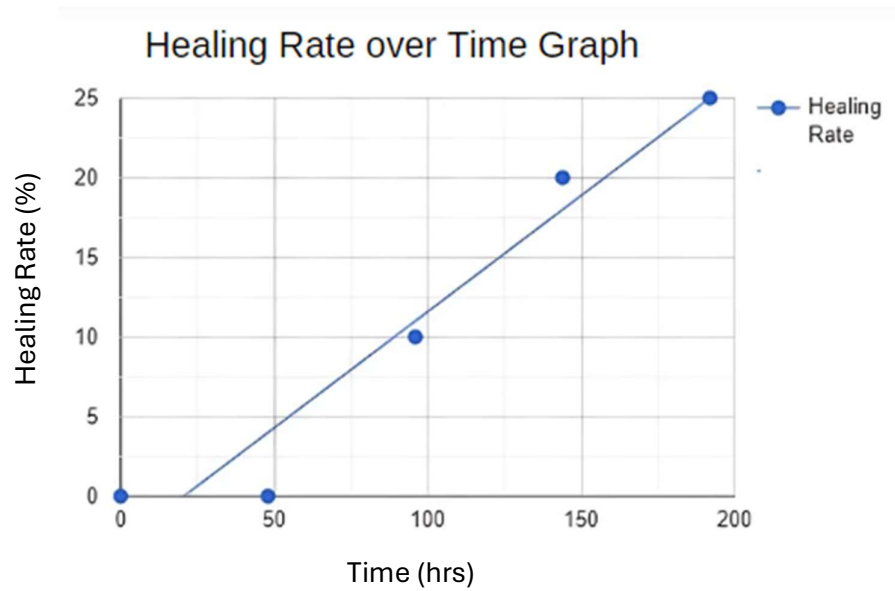
Summary of Key Results:

Parameter	Without Capsules	With Capsules
Crack Width (Final)	6.63 cm	0.36 cm
Healing Rate	0%	88%
Healing Time	No Healing observed	Healing is observed
Overall Outcome	Asphalt degrades and cracks widen	Cracks heal with bacterial action

Crack Width over Time Graph



This graph illustrates the change in crack width over time for both samples. Without capsules, the crack width steadily increases, while with capsules, the crack width decreases as *Bacillus pseudofirmus* bacteria produce calcium carbonate, sealing the cracks.



This graph illustrates the healing rate of self-healing asphalt over time. The linear trend shows a consistent increase in the healing rate as time progresses, indicating the gradual activation and effectiveness of the self-healing mechanism.



Initially



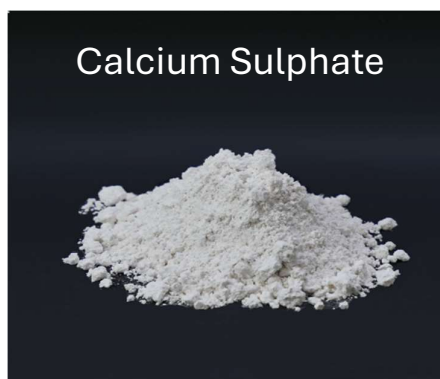
Final Results

Challenges Faced

Throughout the development of Self-Healing Asphalt, we encountered several challenges that required repeated experimentation and adaptation. These challenges often resulted in setbacks but ultimately guided us toward finding the optimal solution. Below are the key difficulties we faced:

Material Selection Issues

- **Initial Choice of Calcium Sulphate:** We initially selected calcium sulphate as the healing agent due to its availability and properties. However, our experiments yielded no significant results, as the material failed to interact effectively with the bacterial cultures for crack healing.
- **Switch to *Bacillus Megaterium*:** To improve performance, we replaced *Bacillus pseudofirmus* with *Bacillus megaterium*, which has been used in other bio-based applications. Unfortunately, the bacteria showed no noticeable activity in the asphalt environment, likely due to incompatibility with the binder and encapsulation method.
- **Testing *Bacillus Subtilis*:** Another alternative we tested was *Bacillus subtilis*, a widely studied bacterium for biological processes. Despite multiple trials, it also failed to produce the desired self-healing effect in the asphalt cracks.



Composition Variations

- **Varying Capsule Content:** We tested different proportions of capsules in the asphalt mix, ranging from 1% to 6% by weight of the binder. While higher proportions caused uneven distribution and material instability, lower proportions failed to deliver effective healing.
- **Adjusting Capsule Ingredients:** Modifications to the gelatin capsules, such as altering calcium lactate concentrations or experimenting with different protective coatings, did not yield significant improvements in early trials.

Experimental Failures

- **Premature Capsule Rupture:** In some trials, capsules ruptured during the mixing process due to excessive heat or mechanical stress, rendering them ineffective for self-healing.
- **Lack of Healing Activity:** In multiple experiments, no visible reduction in crack width was observed, indicating that the selected bacteria and healing agents were either inactive or unsuitable under the test conditions.

Trial-and-Error Approach

- The project demanded numerous iterations of testing and adjustments. Each failed attempt required revisiting the formulation, refining the encapsulation process, or modifying the experimental setup. This trial-and-error approach consumed significant time and resources.

These repeated failures highlighted the complexity of designing an effective self-healing asphalt system. However, they also reinforced the importance of persistence and adaptability.

Conclusion

The Tables and Crack Width Over Time Graph clearly demonstrate the difference in performance between asphalt with and without self-healing capsules. The data shows that asphalt without capsules exhibits a continuous increase in crack width over time, with no signs of healing. In contrast, asphalt with capsules successfully reduces crack width, showcasing the ability of *Bacillus pseudofirmus* bacteria to heal the cracks by producing calcium carbonate.

The crack width over time graph visually reinforces these findings, with the untreated asphalt's crack width increasing steadily, while the self-healing asphalt's crack width decreases, reflecting the healing process in action.

These results confirm the effectiveness of self-healing asphalt and highlight its potential for improving the durability and sustainability of road infrastructure by significantly extending the lifespan of asphalt surfaces through bacterial crack repair.